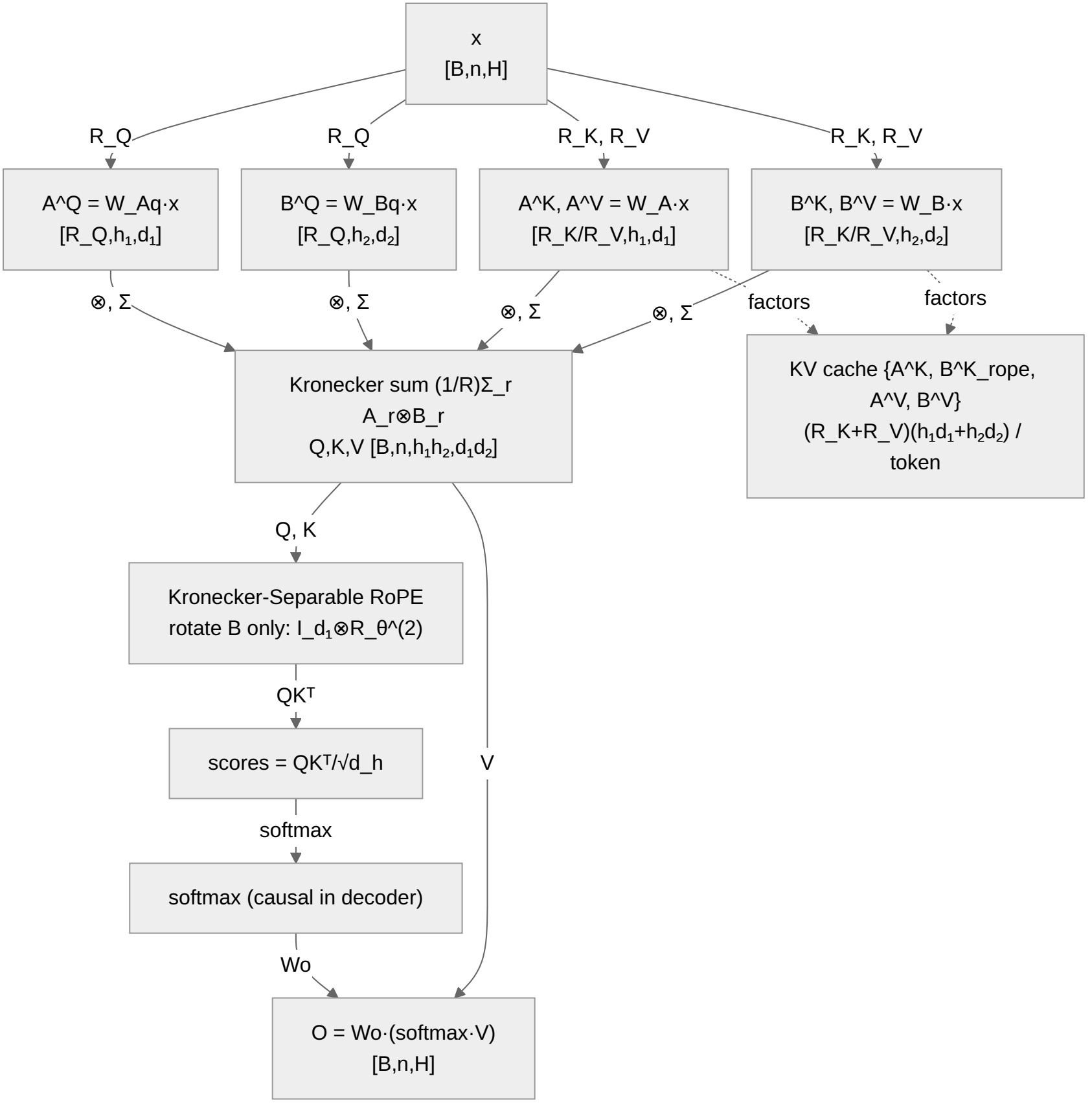
x
[B,n,H]
R_Q
R_Q
R_K, R_V
R_K, R_V
A^Q = W_Aq·x
[R_Q,h1,d1]
B^Q = W_Bq·x
[R_Q,h2,d2]
A^K, A^V = W_A·x
[R_K/R_V,h1,d1]
B^K, B^V = W_B·x
[R_K/R_V,h2,d2]
⊗, Σ
⊗, Σ
⊗, Σ
⊗, Σ
factors
factors
Kronecker sum (1/R)Σ_r
A_r⊗B_r
Q,K,V [B,n,h1h2,d1d2]
KV cache {A^K, B^K_rope,
A^V, B^V}
(R_K+R_V)(h1d1+h2d2) /
token
Q, K
V
Kronecker-Separable RoPE
rotate B only: I_d1⊗R_θ^(2)
QKᵀ
scores = QKᵀ/√d_h
softmax
softmax (causal in decoder)
Wo
O = Wo·(softmax·V)
[B,n,H]